

Power from the Industrial Revolution

Coal which was the driver of the Industrial Revolution in the 18th and 19th centuries continues to be a major source of power until today. As of December 2015, there are nine coal-fired plants operating in the Philippines with a combined output of approximately 4200 MW of power. The management of one of the coal-fired electric power generating plants is studying the plant's operational set-up in order to comply with the latest emission standards under the air pollution control laws. For the plant in question, the maximum emission rates are:

- a. Maximum sulfur oxide emission: 3000 (parts per million) PPM
- b. Maximum particulate emission (smoke): 12 kg/hour

Coal is brought to the plant by barge and dumped onto stockpiles near the plant. From there, it is carried by a conveyor belt to the pulverizer unit where it is pulverized and fed directly into the combustion chamber at the desired rate. The heat produced in the combustion chamber is used to make steam to drive the turbines.

Two types of coal are used at the plant: Grade A which is hard and clean burning coal with a low sulfur content and fairly expensive; Grade B which is a cheap relatively soft and smoky coal with high sulfur content. The table below shows the pollutant emission properties of the two types of coal.

Coal	Sulfur Oxides in Flue Gases(PPM)	Particulate Emission/Ton
A	1800	0.5kg
B	3800	1.0kg

The thermal value in terms of steam produced is higher for Coal A than for Coal B, namely 24000 pounds per ton for A against 20000 pounds per ton for B. Since Coal A is a hard coal, the pulverizer unit can handle at most 16 tons of Coal A per hour, whereas it can pulverize up to 24 tons of Coal B per hour. The conveyor loading system has a capacity of 20 tons per hour regardless of which coal is loaded.

To enable management to determine the margin of safety available to meet peak demands for power, one of the several questions management wants to address is: Given the constraints on emission of pollutants and given the grades of coal available, what is the maximum output of electricity of the plant?